A Typed Ledger for Noisy Spectral Tails, Graded Counterloops, and Their Missing Glue

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Abstract

The variable-rank spectral tail of RH-282 fills one entry of the noisy branch, but it cannot be transferred silently to the monodromy counterloop branch. We update the two five-obligation ledgers and add an explicit cross-branch glue bit. The noisy modulus-spectral vector is $(1, 0, 1, 1, 1)$; the graded monodromy vector is $(1, 1, 0, 1, 1)$; the weighted glue bit is zero. Each branch therefore has score four, with complete count zero. Their coordinatewise maximum is the all-ones vector, but that merge is ill typed: the spectral head and graded head are not identified. RH-288 states the exact weighted complement-to-anchor interface needed to legalize the merge. A sufficient decomposition requires both a weighted total-trace bridge and weighted head-to-counterloop transport. Gates A–E remain open.

1 Typed obligations

The five entries are:

1. a legal head;
2. a coefficient bridge to the deterministic numerator;
3. a uniform high-order tail;
4. an analytic target tail;
5. a certified target boundary constant.

The same labels are used for both branches, but the underlying heads are different objects. The noisy branch uses the actual modulus-complete algebraic eigenvalue multiset. The graded branch uses the deterministic finite-radius monodromy counterloop.

branch	head	bridge	tail	target	boundary	score
noisy modulus spectrum	1	0	1	1	1	4
graded monodromy counterloop	1	1	0	1	1	4

2 No coordinatewise merge

Proposition 2.1 (typed-union obstruction)**.** *The coordinatewise maximum of the two vectors is* $(1, 1, 1, 1, 1)$*, but it is not a complete certificate unless the noisy modulus complement satisfies the weighted prefix bridge to the deterministic anchor. By RH-288, it is sufficient to prove on the same clock both the weighted total-noisy-trace bridge to counterloop plus anchor and the weighted noisy-head bridge to the counterloop.*

Proof. The spectral tail estimate controls the complement of the modulus-selected noisy eigenvalues. The graded coefficient bridge subtracts moments of a different finite multiset. Without an equality or asymptotically vanishing weighted complement-to-anchor prefix, the tail of one factor and the prefix of the other do not occur in a common determinant decomposition. Writing total trace as head plus complement, RH-288 decomposes that missing prefix into a total-trace/counterloop/anchor error minus a head/counterloop error. Controlling only the latter leaves the former open. The compound interface is currently false/open, so coordinatewise union changes the object mid-proof. □

3 Status

The cross-branch weighted-glue bit is zero. Both branch-complete counts are therefore zero even though both raw scores equal four. RH-286 improves the finite-radius diagnostic target and RH-287 gives a rate-free growing prefix; neither supplies the two synchronized weighted estimates or the direct complement-to-anchor prefix.

Remark 3.1. No Gate-A determinant identification has been completed. Gates B–E also remain false/open. Nothing here constructs a Hilbert–Polya operator, identifies Riemann zeros, proves a von Mangoldt trace identity, or proves RH.